Description

Image





Caption

1. Knot tied in a polyamide rope. © Brighterorange at en.wikipedia - (CC BY-SA 3.0) 2. Locking nut with polyamide insert to lock its screw in place. © Cav at en.wikipedia - Public domain 3. Students creating Nylon-6,6 in the laboratory at the University

The material

Back in 1945, the war in Europe just ended, the two most prized luxuries were cigarettes and nylons. Nylon (PA) can be drawn to fibers as fine as silk, and was widely used as a substitute for it. Today, newer fibers have eroded its dominance in garment design, but nylon-fiber ropes, and nylon as reinforcement for rubber (in car tires) and other polymers (PTFE, for roofs) remains important. It is used in product design for tough casings, frames and handles, and - reinforced with glass - as bearings gears and other load-bearing parts. There are many grades (Nylon 6, Nylon 66, Nylon 11 etc.) each with slightly different properties.

Compositional summary

(NH(CH2)5CO)n

General properties

Density		1.12e3	-	1.15e3	kg/m^3
Price	*	1.59	-	2.11	GBP/kg
Date first used		1938			

Mechanical properties

Young's modulus		0.94	-	2.04	GPa
Shear modulus	*	0.38	-	0.67	GPa
Bulk modulus		1.2	-	3.26	GPa
Poisson's ratio		0.34	-	0.42	
Yield strength (elastic limit)		39	-	64	MPa
Tensile strength		42	-	72	MPa

Compressive strength		46	-	82	MPa
Elongation		40	-	60	% strain
Hardness - Vickers		12	-	18	HV
Fatigue strength at 10^7 cycles	*	15	-	27	MPa
Fracture toughness	*	3	-	4	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.02	-	0.03	

Thermal properties

Melting point		220	-	260	°C
Glass temperature		44	-	66	°C
Maximum service temperature		90	-	130	°C
Minimum service temperature	*	-80	-	-50	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.21	-	0.28	W/m.°C
Specific heat capacity	*	1.46e3	-	1.6e3	J/kg.°C
Thermal expansion coefficient		110	-	150	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	1.5e19	-	1e21	µohm.cm
Dielectric constant (relative permittivity)		6	-	14	
Dissipation factor (dielectric loss tangent)		0.07	-	0.23	
Dielectric strength (dielectric breakdown)		20	-	40	MV/m

Optical properties

Transparency		Translucent			
Refractive index		1.52	-	1.57	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Acceptable			
Water (salt)		Acceptable			

Soils, acidic (peat)	Acceptable
Soils, alkaline (clay)	Limited use
Wine	Excellent

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Limited use
Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Limited use
Sodium hydroxide (60%)	Unacceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Limited use
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Acceptable

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Limited use
Ethylene glycol	Excellent
Formaldehyde (40%)	Limited use
Glycerol	Acceptable
Methyl alcohol (methanol)	Limited use

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Unacceptable

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Slow-burning

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.69e6			tonne/yr
Reserves, principal component	8.74e7	-	9.66e7	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 7.7	-	8.48	kg/kg
Embodied energy (virgin grade production)	* 120	-	133	MJ/kg
Water usage	* 176	-	194	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 8.24	-	9.12	MJ/kg
Polymer molding (injection, blow), embodied energy	* 26	-	29.6	MJ/kg
Machining, coarse, embodied energy	* 16.7	-	18.7	MJ/kg
Machining, fine, embodied energy	* 21	-	23.8	MJ/kg
Machining, grinding, embodied energy	* 25.3	-	29	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.545	-	0.605	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.87	-	2.14	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.1	-	1.22	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.38	-	1.56	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.66	-	1.9	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 2.54	-	2.8	kg/kg
Embodied energy, recycling	* 22.8	-	25.3	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 30.1	-	31.6	MJ/kg
Combustion CO2	* 2.28	-	2.39	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Nylons have no known toxic effects, although they are not entirely inert biologically. Nylons are oil-derivatives, but this will not disadvantage them in the near future. With refinements in polyolefin catalysis, nylons face stiff competition from less expensive polymers.

Recycle mark



Supporting information

Design guidelines

Nylons are tough, strong and have a low coefficient of friction, with useful properties over a wide range of temperature (-80 to +120 C). They are easy to injection mold, machine and finish, can be thermally or ultrasonically bonded, or joined with epoxy, phenol-formaldehyde or polyester adhesives. Certain grades of nylon can be electroplated allowing metallization, and most accept print well. A blend of PPO/Nylon is used in fenders, exterior body parts. Nylon fibers are strong, tough, elastic and glossy, easily spun into yarns or blended with other materials. Nylons absorb up to 4% water; to prevent dimensional changes, they must be conditioned before molding, allowing them to establishing equilibrium with normal atmospheric humidity. Nylons have poor resistance to strong acids, oxidizing agents and solvents, particularly in transparent grades.

Technical notes

The density, stiffness, strength, ductility and toughness of Nylons all lie near the average for unreinforced polymers. Their thermal conductivities and thermal expansion are a little lower than average. Reinforcement with mineral, glass powder or glass fiber increases the modulus, strength and density. Semi-crystalline nylon is distinguished by a numeric code for the material class indicating the number of carbon atoms between two nitrogen atoms in the molecular chain. The amorphous material is transparent; the semi-crystalline material is opal white.

Typical uses

Light duty gears, bushings, sprockets and bearings, electrical equipment housings, lenses, containers, tanks, tubing, furniture casters, plumbing connections, bicycle wheel covers, ketchup bottles, chairs, toothbrush bristles, handles, food packaging. Nylons are used as hot-melt adhesives for book bindings, as fibers - ropes, fishing line, carpeting, car upholstery and stockings, as aramid fibers - cables, ropes, protective clothing, air filtration bags and electrical insulation.

Tradenames

Adell, Akulon, Albis, Amilan, Ashlene, Capron, Celanese, Chemlon, Durethan, Gapex, Grilon, Grivory, Hylon, Kopa, Latamid, Lubrilon, Magnacomp, Maranyl, Minlon, NSC, Nivionplast, Novamid, Nydur, Nylamid, Nylene, Nypel, Orgamide, Radilon, Schulamid, Selar, Sniamid, Star-C, Star-L, Staramide, Texalon, Ultramid, Vestamid, Wellamid, Zytel

Links

ProcessUniverse

Producers

Reference

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